

ECE4132 assignment 1: feedforward and feedback

Prepared by: Dr. James Saunderson

References and resources

- ‘Getting started with Simulink’ videos from Mathworks: <https://au.mathworks.com/videos/series/getting-started-with-simulink.html>
- K. J. Åström and R. M. Murray *Feedback systems: an introduction for scientists and engineers* Princeton University Press
- G. F. Franklin, D. Powell and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 6th Edition, Pearson and Prentice Hall, 2010.

Introduction

The assignments in ECE4132 will be based on modeling and/or controlling various digital twins of physical systems. In assignment 1, the system we will control is a heater system. A schematic of the system (only showing the heater input and temperature sensor) is shown below.



The heater input is a value between 0 (no input) and 100 (maximum power). The temperature output is in degrees C. The temperature values are quantized and noisy. The heater and the temperature sensor are physically separated by a conducting medium.

The system is also subject to disturbances, such as air movement across the system, which has a (variable) cooling effect. In the Simulink model of the system, there is a disturbance input that we will use to investigate the effect of such disturbances.

We will always assume that the system starts in equilibrium with its environment. This means that initially the temperature sensor shows (a noisy quantized version of) the room temperature.

The assignment focuses on two of the main tasks of control:

1. Making the output of a system behave in a desired way (in this case, regulating the output to be a constant value of 40°)
2. Suppressing the effect of disturbance inputs on the system output.

In this assignment, you will find the step response (output when the input is a step function) of the system, and then compare two very simple strategies that aim to regulate the output temperature at a fixed value:

- **Feedforward control:** This is an open-loop strategy. In this case, the idea is to find an input signal that will cause the recorded temperature to settle at the desired value.

- **Feedback control:** The second strategy is a particular closed-loop control approach called ‘on-off’ or ‘bang-bang’ control. It is based on setting a reference point, and turning the heater on if the measured temperature is below the reference, and turning the heater off if the measured temperature is above the reference.

Each of these strategies will be discussed in more detail later in this document.

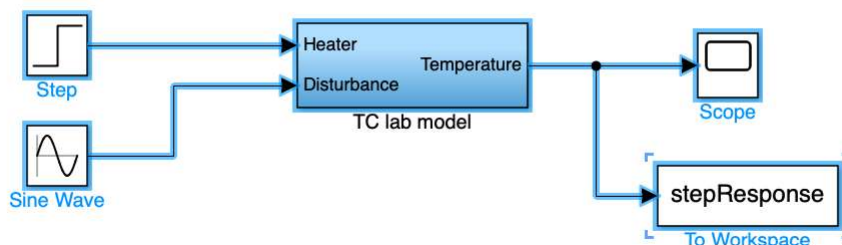
Assessment

This assignment is worth 2% of the final grade for ECE4132. Submit your answers via the Moodle quiz marked **assignment 1**.

1 Step response

In this section you will set up your first Simulink model and find (and record to your MATLAB workspace) the step response of the system.

1. Open MATLAB. Download `studentFiles.zip` from the Moodle page and unzip it. Copy the files to a sensible place and change directory in MATLAB so that your working directory matches the location of the files.
2. From MATLAB, open the Simulink model `TClabsim2016a.slx` (or `TClabsim.slx` if you are using a non-ancient version of Simulink). Note that the model has a subsystem that corresponds to the physical system we want to control.
3. Add a ‘step’, a ‘scope’, a ‘sine wave’ and a ‘to workspace’ block to your model and connect them up as shown below:



A few helpful hints:

- To add a block you can either:
 - double click on the Simulink page and search for the name of the block OR
 - Use the Simulink library browser 
- You can make a branch off a wire by ctrl/cmd-clicking on the wire.
- The ‘scope’ block allows you to plot signals while the simulation is running. It is good for measuring parameters of the signal, but not good for plotting.
- The ‘to workspace’ block takes whatever signal is connected to it when a simulation is running, and saves it to your workspace as a variable. This is very useful for plotting.

4. Set the parameters of the blocks correctly:

- Double click on the 'step' block and adjust the parameters of the step so that it steps from 0 to 100 (the heater values are percentages of full power) at time $t = 0$.
- Double click on the 'sine wave' block and adjust the parameters so that the output is just the constant 0 function. (Later we will use an actual sinusoidal input here, but for now we will just set this input to zero.)
- Set the 'scope' axes to automatically rescale by double clicking on the block and then View→Configuration Properties... and then changing the 'Axes scaling' field to 'auto'.
- Double click on the 'to workspace' block and change the name of the variable to `stepResponse`. Change the 'save format' to 'timeSeries'. After you run a simulation, the variable `stepResponse` should appear in your MATLAB workspace. It is a structure with fields 'time' and 'data'. The signal values are stored in the variable `stepResponse.data`. The corresponding time instants are stored in the variable `stepResponse.time`.

5. Set the simulation time to 600 (i.e., 10 minutes). Do this by changing the box in the top right of the model window to 600. In older versions of Simulink this looks like:



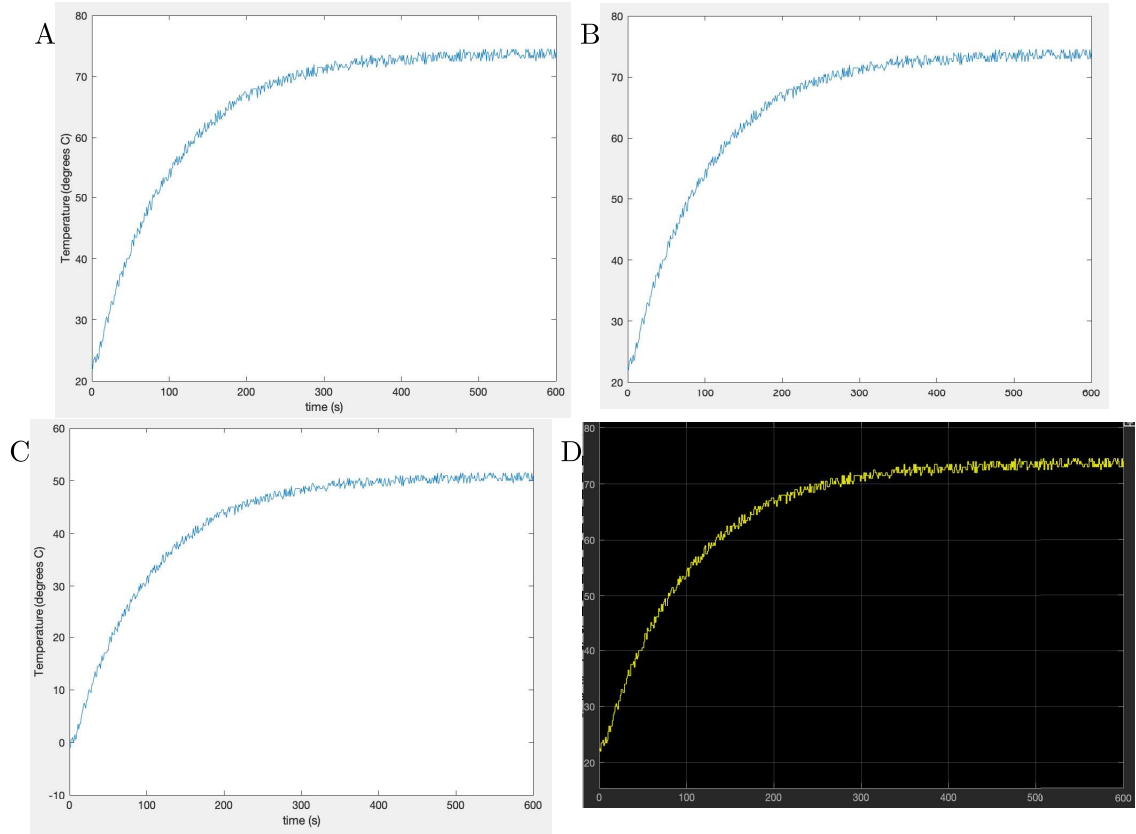
In newer versions of Simulink this looks like



6. You are now ready to find the system step response. Click the 'play' button to run the simulation.
7. Plot the step response of the system in MATLAB. Use the `plot` command together with `stepResponse.Time` and `stepResponse.data`.

Answer the following questions in the assignment 1 Moodle quiz:

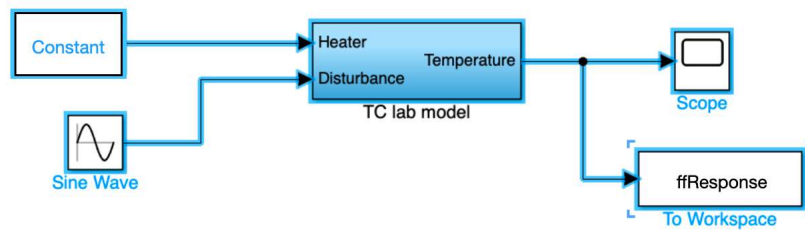
- Q1 The steady-state value of the temperature is in the range ____ ± 2 degrees C. (Fill in the blanks with a positive integer.)
- Q2 The room temperature (in the simulation) is in the range ____ ± 2 degrees C. (Fill in the blanks with a positive integer.)
- Q3 Which of the following is a correct labelled plot of the step response of the system? (Note that it may be the case that none of these exactly match yours, due to the randomness in the simulation, but there should be one that is very similar.)



2 Simple feedforward control

In this section, you will implement a feedforward controller that aims to maintain the `temperature` output at a constant 40° . The basic idea is just to put a (carefully chosen) constant input into `heater` that is just right to maintain the temperature at 40° given that heat is lost due to conduction, convection, and radiation.

The following schematic diagram shows the system and the controller (just a carefully chosen constant input where the constant is chosen to be a value between 0 and 100).



2.1 Modeling

To predict what an appropriate input value might be, we use a very simple model of our system. Let T_{ss} be the steady state temperature when a constant input of H is applied as the first heater input level. Assume that the relationship between the heater input and the difference between the steady-state temperature and the room temperature is linear, i.e.,

$$T_{ss} - T_{room} = C H. \quad (1)$$

where C is some constant. Concretely, this model says that if, for example, we double the heater level, then we expect the difference between steady-state temperature and room temperature

to double. (We will develop more sophisticated dynamic models of our system in upcoming assignments.)

Answer the following question in the assignment 1 Moodle quiz:

- Q4 In Q2 above, you found the value of T_{room} . In Q1 above, you found the value of T_{ss} when $H = 100$. Based on these values, find the value of the constant C in (1). Using your value of C , the value of H that makes $T_{ss} = 40$ is _____. (Round your answer to the nearest integer.)

2.2 Simulation

1. In your SIMULINK model, replace the ‘step’ source block with a ‘constant’ source block (under ‘sources’). You can change the constant value output by this block by double clicking on it and changing the appropriate field. Connect the ‘constant’ block as the input to **heater**. Change the variable name of the ‘to workspace’ block to **ffResponse**.
2. Adjust the value of the constant block manually, and repeatedly re-run the simulation, until the output temperature is steady at (approximately) 40°C. Start with the value you estimated in Section 2.1 above, and adjust it as necessary by re-running the simulation.

Answer the following question in the assignment 1 Moodle quiz:

- Q5 In simulation, what value of the constant block gives a steady-state output temperature of 40°? Round your answer to the nearest integer.
- Q6 There may be a small difference between your predicted heater level in Q4 and the simulated heater level in Q5 needed to regulate the system output at 40°C? Consider the following three potential reasons why you may observe such a difference.
- I The model (1) used for prediction is an oversimplification, and does not fully reflect the mildly non-linear relationship between the steady-state input value and the steady-state output value.
 - II The system response over 600 seconds does not quite reach steady state introducing a systematic error in our estimation of C , and hence our prediction of H .
 - III Since the system temperature starts at the room temperature, rather than at zero, the model (1) does not make sense.

Which of the following is correct?

- A Statement I is the only valid reason.
 - B Statement II is the only valid reason.
 - C Statements I and II are valid reasons, statement III is not a valid reason.
 - D Statements I, II, and III are all valid reasons.
3. Now we will use the sine wave source to add an unmodeled disturbance to the system. Change the parameters of the sine wave so that it is the signal

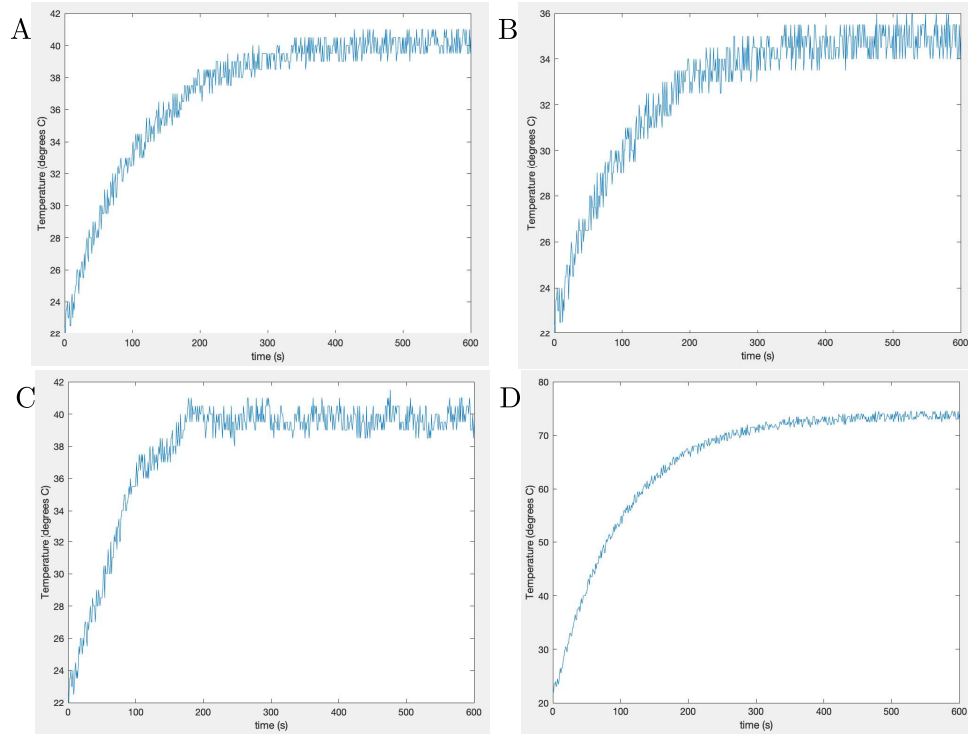
$$H_{\text{dist}}(t) = 10 \sin(2\pi t \times 0.1) + 10. \quad (2)$$

Recall that the disturbance input models cooling effects (like airflow), so the larger the disturbance is the more the system is cooled. A sinusoidal disturbance could model a cooling fan in the room that is changing direction periodically, with period 10 seconds. Every 10 seconds the fan is pointing directly at our heater system (so the cooling is maximized). Note that we did not model this disturbance when coming up with our feedforward control scheme.

4. Run the simulation with this additional disturbance, and plot the system response.

Answer the following questions in the assignment 1 Moodle quiz:

Q7 Which of the following is a correct labeled plot of the measured temperature when you simulate the system with the sinusoidal disturbance?



Q8 Can your feedforward control scheme still maintain the temperature at 40° in the presence of this sinusoidal cooling disturbance?

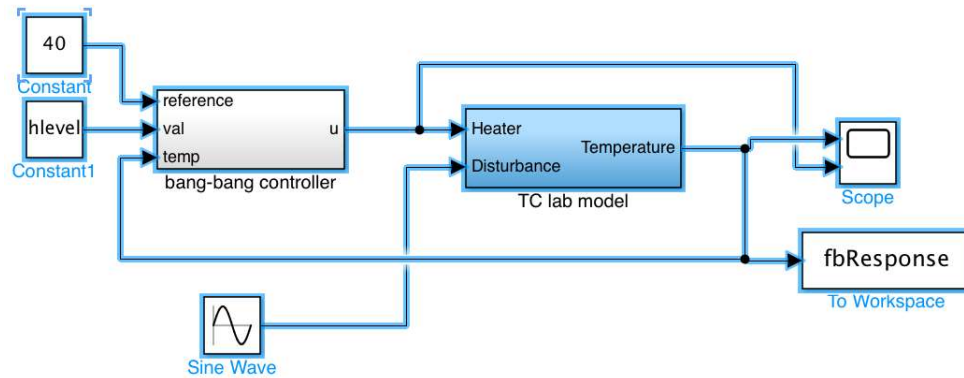
3 ‘Bang-bang’ feedback control

In this section you will implement a very simple feedback control scheme known as bang-bang control. It has this name because the control policy either fully turns on the actuator, or fully turns off the actuator, rather than gradually ramping it up and down. In our case, we will choose to turn on `heater` to some fixed level `val` if `temperature` is below a given threshold `reference`, and fully turn off `heater` if `temperature` is above that same threshold, i.e.,

$$H = \begin{cases} \text{val} & \text{if } \text{temperature} < \text{reference} \\ 0 & \text{otherwise} \end{cases}$$

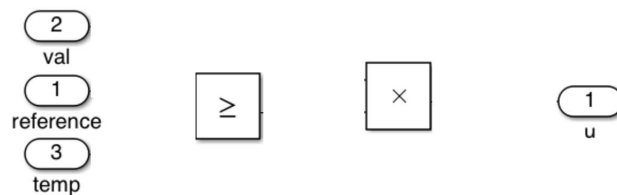
This is a feedback control scheme, because we are choosing our system input as a function of the measured output.

The following schematic diagram shows the system and the controller:



Note the presence of feedback. The details of how the controller is implemented using blocks are not shown. (This is one of the tasks below.)

1. In Simulink, build the block diagram shown above. Set that constant value of the **reference** input to 40, and the constant value of **val** input to 100. Set the parameters of the 'sine wave' block so that the sinusoidal disturbance has amplitude zero and zero bias.
2. Double-click on the bang-bang controller subsystem block to open it. Inside the subsystem block you will implement the controller. To implement the controller, identify the two blocks shown below, add them to your model, and complete the block diagram by correctly wiring up the blocks. (Be careful about the ordering of the inputs!)



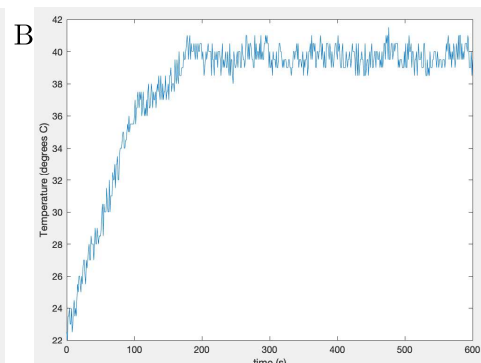
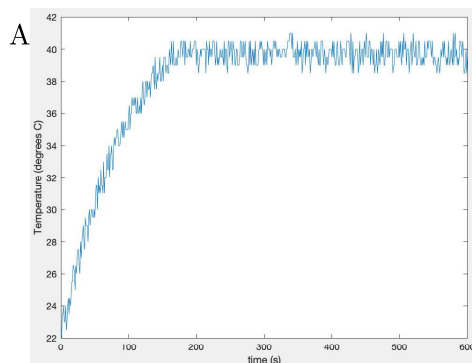
3. Go back to the main system by clicking the up arrow

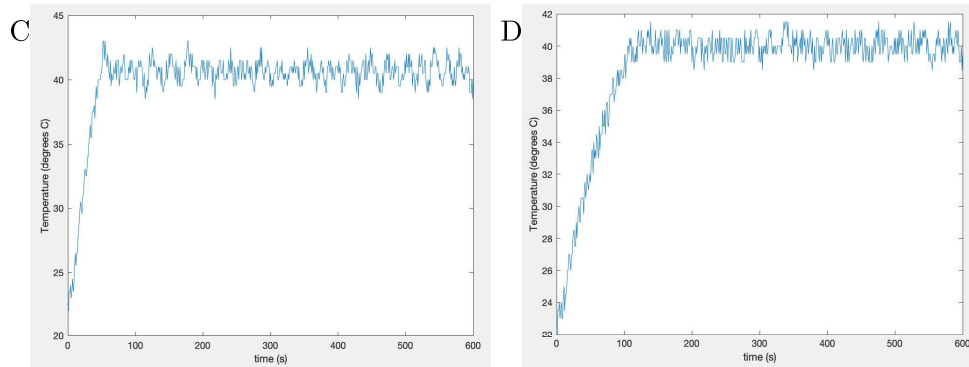


and run the system. Observe the system response on the scope, save it to the workspace, and plot it.

Answer the following questions in the assignment 1 Moodle quiz:

- Q9 Which of the following shows a correctly labelled plot of the temperature output of the system with a feedback controller and $val = 100$?





Q10 Does the feedback system reach steady-state faster or slower than for the feed-forward controller from section 2? Does the feedback system response show more or less overshoot than the feed-forward controller from section 2?

4. Reduce the 'on' state of the bang-bang controller by setting `val = 50` and rerun the simulation.

Answer the following question in the assignment 1 Moodle quiz:

Q11 Does the system response with `val = 50` reach steady-state faster or slower than the system with `val = 100`? Does the system response with `val = 50` show more or less overshoot than the system with `val = 100`?

5. Set the parameters of the sine wave block so that the sinusoidal disturbance is the same as in Equation (2). Rerun the simulation.

Answer the following question in the assignment 1 Moodle quiz:

Q12 Which of the following statements is correct?

- A Unlike the feed-forward controller, the bang-bang controller subsystem uses a model of the disturbance to cancel out the disturbance. That is why the effect of the disturbance is reduced in the presence of the feedback controller.
- B Just like the feed-forward controller, the bang-bang controller subsystem does not use any model of the disturbance. That is why the disturbance is not reduced in the presence of the feedback controller.
- C Just like the feed-forward controller, the bang-bang controller subsystem does not use any model of the disturbance. Despite this it reduces the effect of the disturbance when compared to the feedforward controller. This is because it uses measurements of the system output to adaptively compensate for the effect of the disturbance.
- D Unlike the feed-forward controller, the bang-bang controller subsystem uses a model of the disturbance to cancel out the disturbance. It also reduces the effect of the disturbance by using measurements of the system output to adaptively compensate for the effect of the disturbance.

You have reached the end of the assignment! Please ensure you have answered all the questions on the associated Moodle quiz.